Transfer: circuits and induction

antocciphysics | TEACHER KEY | CPA + MATCHING SEI

Learning target: Apply circuit relationships and explain an induction observation.

Accept equivalent correct methods and clearly labeled models. Language supports do not change the physics demand. Supplied data are examples; actual lab measurements and teacher checkpoints must be recorded from the lesson.

01. Warm-up: bring back an earlier idea *Review: Unit 7, Day 15.*
A generator. Some energy can become heating, friction or sound.

02. Worked example: explain the model
Ammeter in series measures 2 A; voltmeter across resistor measures 12 V; input power 24 W.

03. Read and interpret the model
Both resistor ends share the battery’s two nodes.

04. Apply the physics
2 A and 1 A; total 3 A; 18 W.

05. Explain your reasoning
Motion changes flux, inducing emf; mechanical work supplies energy when a current transfers energy to the circuit.

06. Review: connect an earlier idea *Review: Unit 7, Day 6.*
V=IR=24 V.

07. Exit check
30 J; heat or sound.

08. STRETCH: test the model
Equivalent 6 ohms; same 2 A total and 24 W; each branch 1 A and 12 W.

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01. Warm-up: bring back an earlier idea *Review: Unit 7, Day 15.*
Combined fraction=0.4402 or 44.02%, multiplying the two efficiencies.

02. Worked example: explain the model
Ammeter in series measures 2 A; voltmeter across resistor measures 12 V; input power 24 W.

03. Read and interpret the model
Both resistor ends share the battery’s two nodes.

04. Apply the physics
$R_{parallel}$=4 Ω, total 8 Ω; 1.5 A total; 0.75 A per branch; 18 W.

05. Explain your reasoning
Motion changes flux, inducing emf; mechanical work supplies energy when a current transfers energy to the circuit.

06. Review: connect an earlier idea *Review: Unit 7, Day 6.*
R=40 Ω; V/I is constant at these points under the measured conditions.

07. Exit check
1 V.

08. STRETCH: test the model
Total resistance 4 ohms, current 3 A, total 36 W. Series 18 W; branches 9 W each; sum 36 W.